SV16 to-do list - rebuild step 1

Use the step-1 copy SBSV-16_rebuild_step1_values.kicad_pcb as the starting file. Tick each item when done. Items marked (not bought) need purchase before the board can work.

A. Footprint changes in KiCad (Footprint Editor / Change Footprint)

[ ] C25, C30: change footprint from C_0805 (ceramic) to CP_Elec_4x5.4 (electrolytic), value 10 uF 35 V.

[ ] C32: change footprint from C_0805 to C_0603 (470 nF, spec: 0603).

[ ] R9, R10, R11, R12 (100 ohm): change footprint from R_1206 to R_0805 (bought 100 ohm pack is 0805).

[ ] L1, L2 (10 uH CD54): footprint is L_Sunlord_SWPA6045S. Spec says L_Sunlord_MWSA0518 (5.4x5.2 mm). Confirm the footprint against the bought part datasheet and update.

[ ] J7 (2x25 spare header): add footprint Connector_PinHeader_2.54mm:PinHeader_2x25_P2.54mm_Vertical, mark DNP / not fitted.

[ ] D6 (red LED, INITN): mark DNP (spec: do not fit by default).

[ ] FB1 (600 ohm ferrite, 0603): no ferrite bought. Either buy one, or replace the footprint with a solder bridge (0 ohm jumper pads).

[ ] R21, R22, R23, R24 (0 ohm): no 0 ohm bought. Use solder-bridge pads. Spec: populate one pair only (R21/R23 OR R22/R24).

[ ] R5 and R25 (4.7k): change footprint from R_0603 to R_0805, and use the 4.7k 0805 pack you bought.

[ ] R6, R7, R8 (1k): change footprint from R_0603 to R_1206, and use the 1k 1206 pack you bought.

[ ] Y2 (12 MHz): change footprint from Crystal_HC49-4H (through-hole) to the SMD HC49/US footprint for the bought 12 MHz surface-mount crystal (2-pin, passive, OK for CH340G).

[ ] C36, C37 (22 pF): change footprint from C_0603 to C_0805, to match the YAGEO CC0805JRNPO9BN220 (22 pF C0G 0805 50 V 5%) you are buying.

[ ] R47 (10k, 1%): change footprint from R_1206 to R_0603, and buy the 1% 0603 part (YAGEO RC0603FR-0710KL).

B. Value decisions still open (you decide, then set the value in KiCad)

[ ] R44: no value in spec. Set the value, or leave DNP.

[ ] R45: 100k on board, not in spec. Keep or remove.

[ ] R48, R49: AP62300 EN divider from VM_IN. Spec gives no values. Board has 100k (R48) and 33k (R49). Confirm against the AP62300 datasheet.

[ ] C34: 100 nF on board, not in spec. Keep or remove.

[ ] C35: 100 pF on board, not in spec. Keep or remove.

C. Values already changed in the step-1 copy (verify in KiCad)

[ ] Q2: BC817 changed to MMBT3904. Q4: BC807 changed to MMBT3906 (SOT-23, same pinout per spec).

[ ] C36, C37: 100 nF changed to 22 pF (CH340G crystal load caps, spec).

[ ] Resistors: values set per spec (see BOM). R1, R13-R19, R26, R32, R47 = 10k; R2-R5, R25 = 4.7k; R6-R8, R31, R33, R34 = 1k; R9-R12 = 100 ohm; R27-R30 = 470 ohm; R36, R37 = 5.1k; R38 = 12.4k; R39, R46 = 33k; R40, R41, R43 = 100k.

[ ] Check the file opens with all 141 footprints and that the tracks are unchanged (this copy was checked: 141 footprints, 1,783 segments, 179 vias).

D. Schematic / design checks from the spec (must be fixed in the schematic)

[ ] U5 (AP62300) pin 4 (FB) must be connected to 3V3. Spec says the earlier sheet left it floating.

[ ] U6 (MP1584): add the COMP RC compensation network, values from the datasheet.

[ ] U6 feedback divider: R38 = 12.4k top, R39 = 33k bottom (already set). Keep R39 below 40k.

[ ] C23 (470 uF 16 V) on a 12 V input: 16 V is 1.33x derated. Consider a 25-35 V part if you can still change the order.

[ ] Oscillator Y1: active 25 MHz XO (spec). Confirm the pinout (VCC, GND, OUT, EN) against the bought part before routing.

[ ] Y2 (12 MHz): check the footprint against the bought crystal (HC49-4H vs 3225 SMD).

[ ] LP5907 output cap (C42): electrolytic is used here, spec wants low-ESR ceramic. Check stability on the bench.

E. Purchases still needed (not in any cart screenshot)

[ ] U1: FPGA LFE5U-12F-6TG144C (not bought).

[ ] U3: 1 more W25Q64JVSSIQ (you have 1, need 2).

[ ] Y1: active 25 MHz XO 3.2x2.5 4-pad (YIC OSC25M-3.3I/S3-25T or equivalent). The 3 bought 2-pin 25 MHz crystals cannot be used here.

[ ] Y2: 12 MHz SMD crystal HC49/U (surface mount, 2-pin). To buy. Footprint change needed (see A).

[ ] D8/D9/D11: 1 more SS34 (you have 2, need 3).

[ ] 10k 0603 5% resistors: YAGEO RC0603JR-0710KL, 11 pieces (R1, R13-R19, R26, R32). To buy.

[ ] 10k 0603 1% resistor: YAGEO RC0603FR-0710KL (1%, 0603), 1 piece for R47. To buy. Use this 1% part, not the 5% RC0603JR part, because the spec asks for 1%.

[ ] 22 pF 0805 C0G caps: YAGEO CC0805JRNPO9BN220 (22 pF, 50 V, 0805, 5%), 2 pieces (C36, C37). To buy.

[ ] 4.7k (R5, R25) and 1k (R6-R8): use the 0805 and 1206 packs you already bought, with the footprint changes in section A. No purchase needed.

[ ] Your own list: barrel jack J9, SW1/SW2, headers J1-J6, J10, JP1 shunt.

F. Routing and checks (after A-D are done)

[ ] Re-route the board from the updated placement. Tracks in this file are not verified.

[ ] Run DRC: 0 unconnected, 0 shorts, 0 clearance, 0 edge or courtyard errors. Zones filled.

[ ] Only after clean DRC: generate Gerber, drill, BOM and position files.

[ ] Do not send the board to fabrication until all of the above are done.